

emf induced in it due to the decreasing magnetic flux linkage.)

24. Ans: B

Using the right-hand grip rule, the current in the vertical wire produces a magnetic field which is directed into the plane of the paper at the location of the loop. Hence, this produces a magnetic flux linkage Φ with the loop. As the loop moves to the right, Φ decreases. By Faraday's law, an emf is induced in the loop and an induced current flows

in the loop because it is a closed circuit. By Lenz's law, the current flows in such a direction so as to oppose the decrease in Φ . Using the right-hand grip rule, the current flows clockwise around the loop, so as to produce a magnetic field which is directed into the plane of the paper (within the loop).

Also, by Lenz's law, the induced current will produce a magnetic force which opposes the motion of the loop. So the net magnetic force on the loop will be to the left.